

Task 1: Display the current IP address and network interfaces

Run:

- ifconfig

Task 2: Configure a Static IP using nmcli

Check available connections

- nmcli connection show

Configure Static IP

- sudo nmcli connection modify "Wired connection 1" \
- ipv4.addresses 192.168.1.50/24 \
- ipv4.gateway 192.168.1.1 \
- ipv4.dns "8.8.8.8 8.8.4.4" \
- ipv4.method manual

Restart the connection

- sudo nmcli connection down "Wired connection 1"
- sudo nmcli connection up "Wired connection 1"

Verify :

- ip addr

Task 3: Difference Between DNS and DHCP

DNS	DHCP
Converts domain names to IP addresses	Assigns IP addresses automatically
Used for website access	Used for network configuration
Example: www.flipkart.com	Example: Laptop getting IP from Wi-Fi router

Task 4: Configure Firewall using firewalld (RHEL)

Start firewall

- sudo systemctl enable --now firewalld

Check status

- `sudo systemctl status firewalld`

Add

- `sudo firewall-cmd --permanent --add-service=http`
- `sudo firewall-cmd --reload`
- `sudo firewall-cmd --list-services sudo firewall-cmd --list-all`

Task 5: Block outgoing traffic to a specific website

First find the IP address.

- `ping spotify.com`

Block it:

- `sudo firewall-cmd --permanent --add-rich-rule='rule family="ipv4" destination address="YOUR_IP" reject'`

Reload firewall:

- `sudo firewall-cmd --reload`

Test:

- `ping YOUR_IP`